

Lista de Integração

Universidade Federal da Fronteira Sul – UFFS

Curso: Ciência da Computação (2ª fase)

Disciplina: Cálculo 1

Professor: Milton Kist

Sugestões de exercícios. Faça quantos julgar necessário.

Livro: **Cálculo A**: Funções, limite, derivação e integração, 6ª edição. (Flemming, D. M. & Gonçalves, M. B)

Lista 6.2 (pg. 246): 1; 4; 7; 9; 10; 17; 36.

Lista 6.4 (pg. 250): 1; 2; 3; 6; 8; 9; 10*; 15; 18; 23; 30*; 34; 36; 37; 42*; 43.

Lista 6.6 (pg. 255): Todas as questões da 1 até a 10; além disso as questões: 19*; 21; 24*; 26; 36 .

Lista 6.11 (pg. 269): 1; 5; das questões 12 até 34 fazer as pares.

Lista 6.13 (pg. 278): 1; 2; 3; 4; 7*; 9; 11; 13; 17; 19; 21*; 22; 28; 29; 30*.

Lista 6.15 (pg. 290): 1; 2; 4; 5; 7*; 12*; 14 (a, b, f).

Os exercícios indicados com (*) deverão ser entregues.

Bom trabalho.

Lista de Integrales

6.4

10) $\int \sin^4 x \cos x \, dx$

$$\int u^4 \, du = \int u^4 \cdot \cos x \cdot \frac{du}{\cos x}$$

$$\int u^4 \cdot du = \frac{u^5}{5} + C$$

$$\frac{\sin^5 x + C}{5}$$

$$u = \sin x$$

$$\frac{du}{dx} = \cos x$$

$$dx = \frac{du}{\cos x}$$

30) $\int x^{x^2+1} \, dx$

$$\int u^u \, du$$

$$\int x^u \cdot \frac{du}{dx}$$

$$u = x^2 + 1$$

$$\frac{du}{dx} = 2x$$

$$dx = \frac{du}{2x}$$

$$\frac{du}{2x}$$

$$\int x^u \cdot \frac{du}{2x}$$

$$\frac{1}{2} \cdot \frac{x^u}{\ln x} + C$$

$$\frac{1}{2} \cdot \frac{x^u}{\ln x} + C$$

$$\frac{x^{x^2+1}}{2 \ln x} + C$$

$$\frac{2 \ln x}{x^{x^2+1}} + C$$

$$\frac{x^{x^2+1}}{2 \ln x} + C \rightarrow \frac{x^{x^2+1}}{2 \ln x} + C$$

42) $\int \frac{4x^2 \sqrt{6x^3+5}}{x^2(6x^3+5)^{3/2}} \, dx$

$$u = 6x^3+5$$

$$\frac{du}{dx} = 18x^2$$

$$dx = \frac{du}{18x^2}$$

$$\frac{du}{18x^2}$$

$$\int u^{1/2} \, du$$

$$\frac{4}{9} \int u^{1/2} \cdot \frac{du}{18x^2}$$

$$\frac{4}{9} \int u^{1/2} \cdot du = \frac{4}{9} \cdot \frac{u^{3/2}}{3/2} = \frac{4}{9} \cdot \frac{2}{3} \cdot u^{3/2} = \frac{8}{27} (6x^3+5)^{3/2} + C$$

6.6

$$19) \int (x-1)e^{-x} dx$$

$$\int (x-1)e^{-x} dx = (x-1)(-e^{-x}) - \int (-e^{-x}) dx$$

$$= e^{-x}(-x+1) + (-e^{-x})$$

$$= e^{-x}(-x+1-1)$$

$$= -xe^{-x} + e^{-x}$$

$$u = x-1$$

$$du = e^{-x} dx$$

$$du = 1$$

$$dx$$

$$du = dx$$

$$\int du = \int e^{-x} dx$$

$$u = -e^{-x}$$

$$24) \int e^{3x} \cos 4x dx \rightarrow \int e^{3x} \cos 4x dx$$

$$dv = -4 \sin 4x dx$$

$$v = \frac{e^{3x}}{3}$$

$$u = \cos 4x$$

$$v = \cos 4x$$

$$\cos 4x \cdot \frac{e^{3x}}{3} - \int \frac{e^{3x}}{3} (-4 \sin 4x) dx$$

$$= \cos 4x \cdot \frac{e^{3x}}{3} - 1 \cdot \left(\frac{1}{3} \right) \cdot 4 \cdot \int e^{3x} \sin 4x dx$$

$$= \cos 4x \cdot \frac{e^{3x}}{3} + \frac{4}{3} \left(\sin 4x \cdot \frac{e^{3x}}{3} - \int \frac{e^{3x}}{3} \cos 4x dx \right)$$

$$= \frac{\cos 4x \cdot e^{3x}}{3} + \frac{4 \sin 4x \cdot e^{3x}}{9} - \frac{16}{9} \int e^{3x} \cos 4x dx$$

$$\int e^{3x} \cos 4x dx + \frac{16}{9} \int e^{3x} \cos 4x dx$$

$$\frac{\cos 4x \cdot e^{3x}}{3} + \frac{4 \sin 4x \cdot e^{3x}}{9}$$

$$\frac{25}{9} \int e^{3x} \cos 4x dx = \frac{\cos 4x \cdot e^{3x}}{3} + \frac{4 \sin 4x \cdot e^{3x}}{9}$$

$$\int e^{3x} \cos 4x dx = \frac{3 \cos 4x \cdot e^{3x}}{25} + \frac{4 \sin 4x \cdot e^{3x}}{25}$$

$$\frac{3 \cos 4x \cdot e^{3x}}{25} + \frac{4 \sin 4x \cdot e^{3x}}{25}$$

6B

7) $x = y^2, y - x = 2, y = -2, y = 3$

$x = y^2$ $A = \int_2^3 (-y^2 - 1 + 2) dy$

$y = y^2$

$y^2 = y - 2$

$y^2 = -1 + 2 = 1$

$A = \left[\frac{y^3}{3} - \frac{y^2}{2} + 2y \right]_{-2}^3$

$A = \left(\frac{27}{3} - \frac{9}{2} + 6 \right) - \left(\frac{-8}{3} - \frac{4}{2} - 4 \right)$

$A = \frac{14 - 9 + 12}{2} - \left(\frac{-4 - 6 - 12}{3} \right)$

$A = \frac{21}{2} + \frac{26}{3} = \frac{63 + 52}{6} = \frac{115}{6}$

21) $y = -1 - x^2, y = -2x - 4$

$\frac{1}{x+1}, y = 1, 1 - 1 = 0, 2 - 2(1) = 0, 2 - 2(3) = -4$

Descomponível em $x = 1$

usando

$1/x$ tem assintota em $x = 0$

$2x+1$ crescente

-3 vertical

Integral

$\int_3^{-1} \left(\frac{1}{x} - (2x+1) \right) dx$

$A = \int_3^{-1} \frac{1}{x} dx = \ln|x| \Big|_3^{-1} = (\ln(1) - \ln(3))$

$0 - \ln(3) = -\ln(3)$

$A_2 = \int_3^{-1} 2x dx = 2 \left(\frac{x^2}{2} \right) \Big|_3^{-1} = x^2 \Big|_3^{-1} = (-1)^2 - (3)^2 = 1 - 9 = -8$

$1 - y = 2x + 1$

x

$1 = 2x + 1 = x(2x + 1)$

x

$2x^2 + x - 1 = 0$

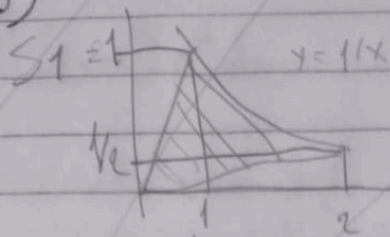
$\frac{-1 \pm \sqrt{1^2 - 4(2)(-1)}}{2(2)} = \frac{-1 \pm \sqrt{17}}{4}$

$A_3 = \int_3^{-1} \frac{1}{x} dx = \ln|x| \Big|_3^{-1} = (\ln(1) - \ln(3)) = -\ln(3)$

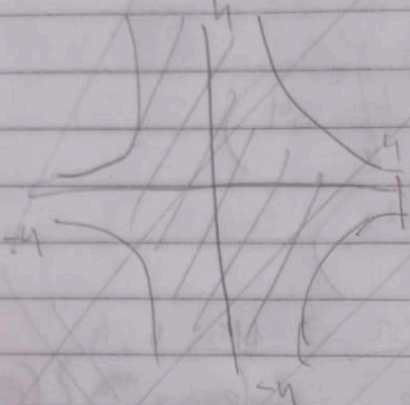
$A_4 = -\ln(3) - (-8) - 2 = \ln(3) + 6 - 2 = \ln(3) + 4$

$(6 - \ln(3))$

30)



$$S = \int_1^2 \left(\frac{1}{x} - \frac{x}{4} \right) dx = \ln x - \frac{x^2}{8} \Big|_1^2$$



$$\int_{-1}^1 \frac{1}{x} dx$$

$$x^2 - 1 = 0 \Rightarrow x = 1, x = -1$$

6.15

$$7) \int_{-\infty}^a e^{5x} dx = \lim_{a \rightarrow \infty} \int_a^{\infty} e^{5x} dx$$

$$\int e^{5x} = \frac{1}{5} e^{5x} + C \Rightarrow \int_a^{\infty} e^{5x} dx = \left[\frac{1}{5} e^{5x} \right]_a^{\infty} = \frac{1}{5} (e^{\infty} - e^{5a})$$

$$\Rightarrow \frac{1}{5} (1 - e^{5a})$$

$$\lim_{a \rightarrow \infty} \frac{1}{5} (1 - e^{5a})$$

$$\frac{1}{5} (1 - 0)$$

$$\frac{1}{5} (1) = \frac{1}{5}$$